

Fixed Features for Exact Two-Layer Logistic SGD

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Abstract

Does distribution-independent learning by exact Gaussian-initialized logistic SGD imply a small common feature representation? We prove a probabilistic-dimension bound for the specified bias-free two-layer process when the step size is at most $1/m$ and the width satisfies $m \geq 16n^2T^2/\delta^2$. Every class learned to expected classification error ε under all input distributions then has probabilistic expected-error dimension at most mn , at error $\varepsilon + \delta$. Both layers train, gradients are exact, and prediction uses the prescribed tail sum. No output-margin assumption is needed. The proof bounds each row's movement for every labeled training sequence. Gaussian anti-concentration gives an initialization-defined exceptional set of small expected mass under any fixed marginal. Outside that set, the complete tail score is exactly linear in mn initial gated-coordinate features, with all products of trained weights retained. Taking $\delta = \varepsilon$ gives $\text{pdc}_{2\varepsilon}(\mathcal{H}) < S$, where $S = m(n+1)$. The result concerns a restricted regime and probabilistic dimension; ordinary dimension is not bounded here.

Keywords: Stochastic gradient descent; probabilistic dimension; random features; ReLU networks.

1. Introduction

Can a neural network learn under every input distribution while requiring a much larger common linear representation? Feldman, Kamath, and Srebro pose this question for a prescribed Gaussian-initialized training process (Feldman et al., 2026). Its two-layer Boolean-cube version uses exact one-example logistic SGD, trains both layers, and predicts with the sum of the scores over the second half of the trajectory. The original representation measure is ordinary dimension complexity, or sign-rank.

We give an affirmative implication for probabilistic expected-error dimension in an explicit small-step regime. Let n be the input dimension, m the width, and T the number of updates. For any $0 < \delta < 1$, the conditions

$$\eta \leq \frac{1}{m}, \quad m \geq \frac{16n^2T^2}{\delta^2} \quad (1)$$

imply that distribution-independent error ε for the exact learner gives

$$\text{pdc}_{\varepsilon+\delta}(\mathcal{H}) \leq mn < S, \quad S = m(n+1). \quad (2)$$

The random feature law is explicit and depends only on n, m .

The reason this statement can avoid an output margin is simple. Once the gates at a test point retain their initial signs, a two-layer score is exactly linear in the corresponding gated coordinates. Training changes the coefficients, including their bilinear dependence on both layers. We keep those coefficients intact. Classification disagreement can therefore

occur only on the set where some gate might change; there is no numerical score error to compare with a margin.

It remains to control that set without imposing assumptions on the marginal. We bound row movement uniformly over all possible labeled training sequences. A Gaussian small-ball calculation then bounds the probability that any initial gate at a fixed point lies within its movement radius. Averaging over an arbitrary fixed marginal gives the required exceptional mass. The marginal may be concentrated on a small support and need not possess a density.

Relation to earlier work. Small parameter movement and fixed-feature behavior are established themes in lazy-training analysis (Chizat et al., 2019). Frozen-gate models retaining products of trained matrices also appear in Allen-Zhu et al. (2019, Section 6.2). Ji and Telgarsky use the same gated-coordinate geometry and Gaussian slab estimates in their logistic-SGD analysis (Ji and Telgarsky, 2020). Their model fixes the output weights and assumes separability in a tangent feature space. Daniely’s conjugate-kernel guarantees also use comparator-dependent complexity (Daniely, 2017). Here the premise is the achieved classification error of the given both-layer process, with no comparator or output-margin hypothesis. Our contribution is the resulting finite-parameter, all-marginal dimension implication; the frozen-gate principle itself is standard.

The connection between gradient-descent learnability and simpler approximating classes is studied by Malach et al. (2021). Their population-gradient theorem has a local regularity assumption and does not directly cover single-example ReLU SGD; see their Remarks 8 and 10. Distributional qualifications also matter even for a single neuron (Yehudai and Shamir, 2020). We work with the probabilistic dimension of Kamath et al. (2020), whose separation from ordinary dimension prevents interpreting (2) as an ordinary sign-rank bound.

2. The process and the representation measure

Let $\mathcal{X}_n = \{-1, 1\}^n$ and $\mathcal{H} \subseteq \{-1, 1\}^{\mathcal{X}_n}$. Throughout, $\text{sign}(z) = 1$ for $z \geq 0$ and $\text{sign}(z) = -1$ for $z < 0$. Write $\text{err}_D(g, h) = \Pr_{x \sim D}[g(x) \neq h(x)]$.

Fix integers $n, m, T \geq 1$ and a step size $\eta > 0$ before choosing h, D . Initialize independently

$$(W_0)_{ji} \sim N(0, 1/n), \quad (a_0)_j \sim N(0, 1/m), \quad f_t(x) = a_t^\top \text{ReLU}(W_t x). \quad (3)$$

For $t = 0, \dots, T-1$, draw $x_t \sim D$ independently, put $y_t = h(x_t)$, and set

$$\begin{aligned} c_t &= \frac{\eta y_t}{1 + \exp(y_t f_t(x_t))}, & v_t &= \mathbf{1}\{W_t x_t > 0\}, \\ a_{t+1} &= a_t + c_t \text{ReLU}(W_t x_t), & W_{t+1} &= W_t + c_t (a_t \odot v_t) x_t^\top. \end{aligned} \quad (4)$$

Both right sides use the old parameters. The indicator in (4) enforces the derivative-zero convention. Normalize the tail score by its positive length:

$$q = \lfloor T/2 \rfloor + 1, \quad F_\omega(x) = \frac{1}{q} \sum_{t=\lceil T/2 \rceil}^T f_t(x), \quad G_\omega = \text{sign } F_\omega. \quad (5)$$

This normalization leaves the prescribed classifier unchanged. The symbol ω includes initialization and examples.

The learning premise at accuracy ε is

$$\forall h \in \mathcal{H} \forall D \in \Delta(\mathcal{X}_n), \quad \mathbb{E}_\omega \text{err}_D(G_\omega, h) \leq \varepsilon. \quad (6)$$

The same m, T, η are used for every pair (h, D) . They may depend on the class, dimension, and requested accuracy.

For $\alpha \geq 0$, let $\text{pdc}_\alpha(\mathcal{H})$ be the least integer $d \geq 0$ for which one law Q of feature maps $\phi : \mathcal{X}_n \rightarrow \mathbb{R}^d$ satisfies

$$\forall h \in \mathcal{H} \forall D \in \Delta(\mathcal{X}_n), \quad \mathbb{E}_{\phi \sim Q} \inf_{u \in \mathbb{R}^d} \text{err}_D(\text{sign}\langle u, \phi(\cdot) \rangle, h) \leq \alpha. \quad (7)$$

The law precedes both h and D ; the vector u may depend on them and on the realized map. There is no norm bound. We use separate notation from ordinary dimension dc , which requires one deterministic feature map representing every target exactly. Finiteness of the cube makes the infimum in (7) a minimum.

3. The feature reduction

The features preserve the initial gate and allow a separate coefficient for each input coordinate:

$$\phi_{W_0}(x)_{j,i} = x_i \mathbf{1}\{w_{0j}, x\} > 0\}, \quad 1 \leq j \leq m, \quad 1 \leq i \leq n. \quad (8)$$

Thus the dimension is mn , rather than m . These are the initial input-layer tangent features with their nonzero output-weight factors removed. The two sets have the same span almost surely.

Theorem 1 (Small-step dimension implication) *Suppose (6) holds for the exact process (3)–(5), with $0 < \varepsilon < 1/4$. Let $0 < \delta < 1$. If (1) holds, the law in (8) witnesses*

$$\text{pdc}_{\varepsilon+\delta}(\mathcal{H}) \leq mn.$$

In particular, $\eta \leq 1/m$ and $m \geq 16n^2T^2/\varepsilon^2$ imply $\text{pdc}_{2\varepsilon}(\mathcal{H}) \leq mn < S$. No restriction on the targets is imposed.

We prove a quantitative statement about the outputs before using the learning premise. Put $\tau = \eta T \sqrt{n}$. For $0 \leq \tau \leq 1$, define

$$\mathcal{B}_{n,m}(\tau) = \min \left\{ 1, \frac{\frac{2}{\pi} \sqrt{nm} \sinh \tau + \sqrt{\frac{2}{\pi}} m \sqrt{n-1} (\cosh \tau - 1)}{2 - \cosh \tau} \right\}. \quad (9)$$

Theorem 2 (Exact scores outside an initialization-defined set) *Assume $\tau \leq 1$. For every finite initialization (W_0, a_0) , there is a set $B \subseteq \mathcal{X}_n$, determined by that initialization and n, m, T, η , with the following property: for every length- T labeled training sequence in $\mathcal{X}_n \times \{-1, 1\}$, there is a coefficient array $C \in \mathbb{R}^{m \times n}$ such that*

$$F_\omega(x) = \langle C, \phi_{W_0}(x) \rangle \quad (x \notin B). \quad (10)$$

For every fixed marginal D , independent of initialization,

$$\mathbb{E}_{W_0, a_0} D(B) \leq \mathcal{B}_{n,m}(\tau) \leq 2n\eta T \sqrt{m} + n^{3/2} m (\eta T)^2. \quad (11)$$

Consequently, (6) implies

$$\text{pdc}_{\varepsilon+\mathcal{B}_{n,m}(\tau)}(\mathcal{H}) \leq mn. \quad (12)$$

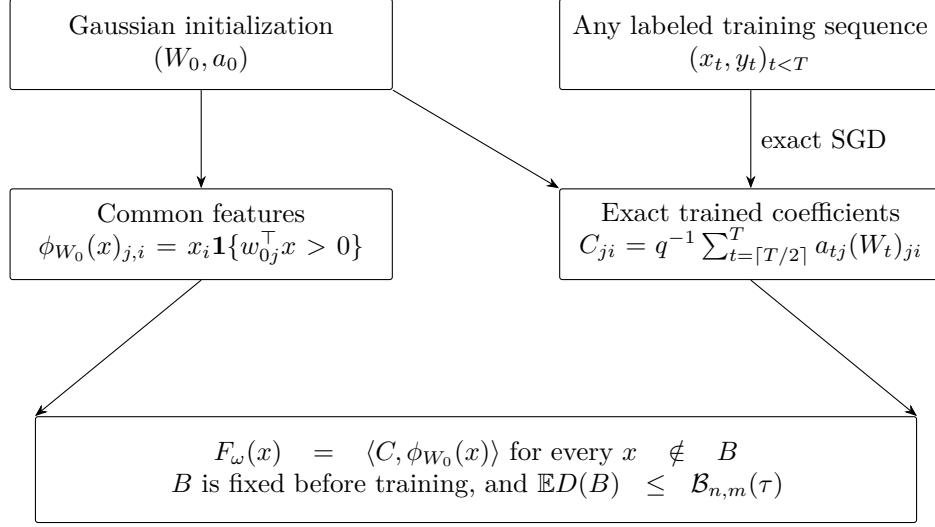


Figure 1: The feature law does not depend on a target or marginal. The full trained products enter only the coefficients. Outside the initialization-defined set B , equality is exact, so no output margin is needed.

The set B works for all training sequences simultaneously, including sequences chosen after seeing the initialization. Only the test-marginal estimate in (11) requires D to be fixed beforehand. It does not assert that $\mathbb{E} \sup_D D(B)$ is small. Figure 1 shows the distinction between the common features and the fitted coefficients.

4. Proof

4.1. A rowwise bound for every training sequence

The two layers feed into each other's movement. Keeping the initial output-weight scale in this comparison is what makes width useful.

Lemma 3 (Row movement) *For every finite initialization and every labeled sequence, each row obeys*

$$|a_{tj}| \leq |a_{0j}| \cosh(t\eta\sqrt{n}) + \|w_{0j}\|_2 \sinh(t\eta\sqrt{n}), \quad (13)$$

$$\|w_{tj} - w_{0j}\|_2 \leq |a_{0j}| \sinh(t\eta\sqrt{n}) + \|w_{0j}\|_2 (\cosh(t\eta\sqrt{n}) - 1). \quad (14)$$

No restriction on η is needed for these bounds.

Proof Fix a row, write $u_t = |a_{tj}|$, $v_t = \|w_{tj}\|_2$, and set $\lambda = \eta\sqrt{n}$. Since $|c_t| \leq \eta$ and $\|x_t\|_2 = \sqrt{n}$, (4) gives

$$\begin{pmatrix} u_{t+1} \\ v_{t+1} \end{pmatrix} \leq \begin{pmatrix} 1 & \lambda \\ \lambda & 1 \end{pmatrix} \begin{pmatrix} u_t \\ v_t \end{pmatrix}$$

coordinatewise. Define the nonnegative coefficients

$$E_t = \sum_{\substack{0 \leq r \leq t \\ r \text{ even}}} \binom{t}{r} \lambda^r, \quad O_t = \sum_{\substack{0 \leq r \leq t \\ r \text{ odd}}} \binom{t}{r} \lambda^r.$$

The matrix power has diagonal entries E_t and off-diagonal entries O_t . As $\binom{t}{r} \leq t^r/r!$, we have $E_t \leq \cosh(t\lambda)$ and $O_t \leq \sinh(t\lambda)$. This proves (13).

For displacement, sum the actual increment norms. The recurrences $O_{t+1} = O_t + \lambda E_t$ and $E_{t+1} = E_t + \lambda O_t$ give

$$\|w_{tj} - w_{0j}\|_2 \leq \lambda \sum_{r < t} u_r \leq O_t u_0 + (E_t - 1)v_0.$$

The hyperbolic bounds prove (14). This argument bounds displacement directly, without subtracting upper bounds on two norms. $\blacksquare$

Set $s = \sinh \tau$ and $b = \cosh \tau - 1$, and define

$$r_j = |a_{0j}|s + \|w_{0j}\|_2 b, \quad B = \bigcup_{j=1}^m \{x \in \mathcal{X}_n : |\langle w_{0j}, x \rangle| \leq \sqrt{n} r_j\}. \quad (15)$$

If $x \notin B$, Cauchy–Schwarz and Lemma 3 imply

$$|\langle w_{tj} - w_{0j}, x \rangle| \leq \sqrt{n} r_j < |\langle w_{0j}, x \rangle| \quad (t \leq T).$$

Every preactivation at x retains its initial, nonzero sign, for every possible training sequence. Zero initial preactivations are included in B .

4.2. A Gaussian estimate with a dependent radius

The radius r_j contains $\|w_{0j}\|_2$, so it is not independent of the initial preactivation. The next calculation accounts for that dependence.

Fix $x \in \mathcal{X}_n$ and a row j . Write $Z = \sqrt{n} w_{0j} \sim N(0, I_n)$ and decompose it along $x/\sqrt{n}$. Then

$$G = \langle w_{0j}, x \rangle \sim N(0, 1), \quad \|Z\|_2 = \sqrt{G^2 + R^2},$$

where R is independent of G and $\mathbb{E}R \leq \sqrt{n-1}$. For $n = 1$, take $R = 0$. Both variables are independent of a_{0j} .

The j th event in (15) implies

$$|G| \leq \sqrt{n} |a_{0j}|s + b\sqrt{G^2 + R^2} \leq \sqrt{n} |a_{0j}|s + b|G| + bR.$$

Since $\tau \leq 1$ gives $b < 1$, this event is contained in

$$\left\{ |G| \leq \frac{\sqrt{n} |a_{0j}|s + bR}{1 - b} \right\}.$$

The bound on the right is now independent of G . The standard Gaussian density is at most $1/\sqrt{2\pi}$, hence

$$\begin{aligned} \Pr[x \text{ belongs to the } j\text{th set in (15)}] &\leq \sqrt{\frac{2}{\pi}} \frac{\sqrt{n} \mathbb{E}|a_{0j}|s + b\mathbb{E}R}{1-b} \\ &\leq \frac{\frac{2}{\pi}\sqrt{n/m}s + \sqrt{\frac{2}{\pi}}\sqrt{n-1}b}{1-b}. \end{aligned} \quad (16)$$

Here $\mathbb{E}|a_{0j}| = \sqrt{2/(\pi m)}$. A union bound over the m rows gives $\Pr[x \in B] \leq \mathcal{B}_{n,m}(\tau)$, uniformly in the fixed point x . Therefore, for any fixed marginal D ,

$$\mathbb{E}D(B) = \mathbb{E}_{x \sim D} \Pr_{W_0, a_0} [x \in B] \leq \mathcal{B}_{n,m}(\tau). \quad (17)$$

There is no union bound over the cube or over training histories.

For the simpler estimate, when $0 \leq \tau \leq 1$ use

$$\sinh \tau \leq \frac{6}{5}\tau, \quad \cosh \tau - 1 \leq \frac{5}{9}\tau^2, \quad 2 - \cosh \tau \geq \frac{4}{9}.$$

These follow from the positive power series and the endpoint bounds. Also $2/\pi < 2/3$ and $\sqrt{2/\pi} < 4/5$. Substitution in (9) gives

$$\mathcal{B}_{n,m}(\tau) \leq 2\sqrt{nm}\tau + m\sqrt{n}\tau^2 = 2n\eta T\sqrt{m} + n^{3/2}m(\eta T)^2. \quad (18)$$

4.3. Exact representation and the distribution quantifiers

Fix an initialization and any labeled training history. Define

$$C_{ji} = \frac{1}{q} \sum_{t=\lceil T/2 \rceil}^T a_{tj}(W_t)_{ji}. \quad (19)$$

For $x \notin B$, the fixed-gate identity gives

$$\begin{aligned} F_\omega(x) &= \frac{1}{q} \sum_t \sum_j a_{tj} \mathbf{1}\{\langle w_{0j}, x \rangle > 0\} \langle w_{tj}, x \rangle \\ &= \sum_{j,i} C_{ji} \phi_{W_0}(x)_{j,i}. \end{aligned}$$

The sum over t is the tail in (5). All bilinear terms are included in C . In particular the two classifiers agree at an output tie outside B .

For any h, D , it follows pathwise that

$$\text{err}_D(\text{sign}\langle C, \phi_{W_0}(\cdot) \rangle, h) \leq \text{err}_D(G_\omega, h) + D(B). \quad (20)$$

For each realized W_0 , the infimum over coefficient vectors is at most the error of *every* array (19), including those selected by different a_0 and training samples. Thus

$$\begin{aligned} \mathbb{E}_{W_0} \inf_u \text{err}_D(\text{sign}\langle u, \phi_{W_0}(\cdot) \rangle, h) \\ \leq \mathbb{E}_{W_0, a_0, x_0, \dots, x_{T-1}} \text{err}_D(\text{sign}\langle C, \phi_{W_0}(\cdot) \rangle, h) \\ \leq \mathbb{E}_\omega \text{err}_D(G_\omega, h) + \mathcal{B}_{n,m}(\tau). \end{aligned}$$

Using (6) proves (12). The feature law is precisely the marginal law of W_0 from (3). No law is reselected when h or D changes. This proves Theorem 2.

Proof [Proof of Theorem 1] Under (1),

$$\tau \leq \frac{T\sqrt{n}}{m} \leq \frac{\delta^2}{16n^{3/2}T} < 1.$$

Equation (18) gives

$$\mathcal{B}_{n,m}(\tau) \leq \frac{2nT}{\sqrt{m}} + \frac{n^{3/2}T^2}{m} \leq \frac{\delta}{2} + \frac{\delta^2}{16\sqrt{n}} \leq \frac{9\delta}{16} < \delta.$$

Apply Theorem 2 and monotonicity in the allowed error. ■

5. The regime and a trajectory-dependent extension

The relevant width-step tradeoff is visible in (18):

$$n\eta T\sqrt{m} \ll 1, \quad n^{3/2}m(\eta T)^2 \ll 1.$$

Increasing width while holding the step size fixed need not satisfy either inequality. With $\eta = 1/m$, however, both terms decrease with width. The sufficient width in Theorem 1 is polynomial in $n, T, 1/\delta$.

Our estimate controls the probability that *some* gate changes at a test point over the entire trajectory. This is stronger than controlling the fraction of changed gates. A bound on a small fraction would usually provide a small score error; without an output margin, that need not bound classification disagreement. The stronger regime is the price paid by this proof for the direct classification transfer.

The scale $\eta = 1/m$ still permits a finite change in the score. In fact, the following limit holds under the exact initialization and one simultaneous update. Its proof is in Appendix A.

Proposition 4 (One-step score scale) *Fix n , $x \in \mathcal{X}_n$, $y \in \{-1, 1\}$, and $0 < \kappa \leq 1$. For width m , take one update on (x, y) with $\eta = \kappa/m$. As $m \rightarrow \infty$,*

$$(f_0(x), f_1(x)) \implies \left(Z, Z + \frac{\kappa y}{2(1 + e^{yZ})} \right), \quad Z \sim N(0, 1/2).$$

This calculation concerns one score increment. It supplies no convergence claim for a target class.

Using the realized logistic factors. A trajectory may move less than the worst-case bound $|c_t| \leq \eta$ predicts. Define its cumulative factor

$$L_T = \sqrt{n} \sum_{t < T} |c_t| = \eta \sqrt{n} \sum_{t < T} \frac{1}{1 + \exp(y_t f_t(x_t))}. \quad (21)$$

Proposition 5 (Cumulative-factor variant) *Suppose (6) holds. If deterministic $0 \leq \lambda \leq 1$ and $0 \leq \rho \leq 1$ satisfy*

$$\forall h \in \mathcal{H} \forall D, \quad \Pr_{\omega}[L_T > \lambda] \leq \rho,$$

then the feature law (8) witnesses

$$\text{pdc}_{\varepsilon+\rho+\mathcal{B}_{n,m}(\lambda)}(\mathcal{H}) \leq mn.$$

Proof Use the comparison matrices $I + \sqrt{n}|c_t|J$, where J swaps the two coordinates. They commute. The coefficient of degree r in their product is at most $L_T^r/r!$, so the proof of Lemma 3 gives the same row-displacement bound with L_T in place of τ . On $\{L_T \leq \lambda\}$, use the set B in (15) constructed from the deterministic value λ . Its expected marginal mass is at most $\mathcal{B}_{n,m}(\lambda)$ by the same Gaussian calculation. The pointwise error inequality (20) now has the additional term $\mathbf{1}\{L_T > \lambda\}$. Average and use its stated uniform probability bound. ■

The extra probability hypothesis is explicit. It is not inferred from the final classification error. Taking $\lambda = \eta T \sqrt{n}$ and $\rho = 0$ recovers the deterministic small-step analysis whenever $\lambda \leq 1$.

6. Discussion

For this regime, the actual updates provide the missing restriction on output selection: outside one initialization-defined set of controlled mass, every possible training history remains in the same mn -dimensional feature span. The proof uses the small initial output weights in the row comparison and the bound on the logistic derivative. It does not require approximating any update.

There are two boundaries to the conclusion. The small-step hypothesis is stronger than the usual requirement that only a small proportion of gates change, and we have not optimized its dependence on n, T . The representation conclusion is also probabilistic and approximate. It does not settle the ordinary-dimension form of Open Question 1, even within this regime. Extending the classification transfer beyond the present gate event would require further control of how score changes interact with the output sign.

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Appendix A. Proof of the one-step score limit

Fix x, y, n as in Proposition 4. Write $z_j = \langle w_{0j}, x \rangle$ and $a_{0j} = b_j / \sqrt{m}$. Then z_j, b_j are independent standard Gaussians. Consequently

$$f_0(x) = \frac{1}{\sqrt{m}} \sum_{j=1}^m b_j \text{ReLU}(z_j) \implies Z \sim N(0, 1/2)$$

by the central limit theorem. Put $Q_m = \sum_j \text{ReLU}(z_j)^2$. The laws of large numbers also give

$$\frac{Q_m}{m} = \frac{1}{m} \sum_j \text{ReLU}(z_j)^2 \longrightarrow \frac{1}{2}, \quad A_m := \sum_j a_{0j}^2 \mathbf{1}\{z_j > 0\} \longrightarrow \frac{1}{2}$$

in probability.

By the fixed-point bound in Theorem 2 with $T = 1$ and $\eta = \kappa/m$, the probability that any gate at x changes during this step tends to zero. On its complement set

$$c = \frac{\kappa y}{m(1 + \exp(yf_0(x)))}.$$

An initially inactive row does not move on this training example. On an active row the simultaneous update is

$$a_{1j} = a_{0j} + cz_j, \quad z_{1j} = z_j + cna_{0j}.$$

The gate remains active on the event under consideration, so multiplying and summing gives the exact identity

$$f_1(x) - f_0(x) = cQ_m + cnA_m + c^2nf_0(x).$$

The final two terms tend to zero in probability, since $|c| \leq \kappa/m$, A_m converges in probability, and $f_0(x)$ is tight. The first term has the joint limit

$$(f_0(x), cQ_m) \implies \left(Z, \frac{\kappa y}{2(1 + \exp(yZ))} \right)$$

by Slutsky's theorem and continuity. An event of vanishing probability cannot affect this convergence in distribution. This proves Proposition 4.

Appendix B. Verification notes

The accompanying executable checks use exact rational arithmetic for simultaneous updates with bounded scalar multipliers $c_t \in [-\eta, \eta]$. That is a larger update family than logistic SGD, and it is the family used by the deterministic comparison proof. The checks enumerate 9,216 bounded-multiplier histories and verify 136,153 exact tail-score identities on points whose gates remain fixed, including output ties. They also check row certificates and zero preactivations. They do not numerically replace the logistic function in the theorem.

A separate numerical grid checks the stated elementary bounds and the simplified width condition. These calculations are diagnostics. The universal claims follow from the proofs above; no convergence experiment is used as evidence for a learning guarantee.